
PH31207 — Classical Mechanics

Tutorial: 01

Date: July 23 2026, 11:00 AM

Problem 1

Consider a reasonably isolated system of three masses m_1, m_2 and m_3 . The radius vector of the i th particle from the origin is $\vec{r}_i$, where $i = 1, 2, 3$. Assume that the interaction between the masses obeys the Newton's law of gravitation. No external force is assumed to be acting on these masses.

1. Evaluate the summation

$$\sum_{i=1}^3 \sum_{\substack{k=1 \\ k \neq i}}^3 \vec{r}_i \times \vec{F}_{ik}$$

explicitly, where $\vec{F}_{ik}$ is the force on the i th mass due to the k th mass.

2. Is the total angular momentum conserved in this case?

Problem 2

Consider

$$\vec{F} = (2xy + z^3) \mathbf{i} + (x^2 + 2y) \mathbf{j} + (3xz^2 - 2) \mathbf{k},$$

where $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ are unit vectors along x , y , and z directions respectively.

1. Find a scalar function $\phi(x, y, z)$ such that $\vec{F} = \vec{\nabla}\phi$. Show explicitly how you obtained ϕ .
2. Calculate $\vec{\nabla} \times \vec{F}$ separately.
3. Is the force conservative?
4. Consider a particle moving under the above force field in 3D. The equation for the trajectory of this particle is given by

$$\vec{r}(t) = 3\mathbf{i} \cos\left(3t + \frac{\pi}{2}\right) + 3\mathbf{j} \sin(4t + \pi) - \mathbf{k}.$$

How much work is done by the force on this particle in the time interval $[0, 2\pi]$?

[Extra: Plot this curve; can you identify it?]